‘Final Submission – Broughty Ferry Media: Customer Renewal Prediction’

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Contents

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1.0 Introduction

This report extends the findings of the checkpoint analysis report for the organisation Broughty Ferry Media by applying the locked Random Forest model to the test dataset consisting of 5,000 customers. The aim of the organisation and this framework remains the same: to maximise the returns from a £20,000 renewal campaign by targeting the customers who are most likely to renew their subscriptions (£40 per contact, £140 renewal revenue, maximum 500 contacts (20,000 / 40)).

The checkpoint report established ten predictor variables for renewal, verified through point-biserial correlation analysis and confirmed by VIF testing. In addition to these predictor variables, it was found that the Random Forest predictive model would generalise the best to unseen data. The data displayed that the Random Forest model had the highest AUC value (0.8903) when compared to the other models (logistic regression = 0.8186, decision tree = 0.8740), as well as the largest amount of profit in the scaled simulation (£6,740 to 150 contacts), making it the best suited for this type of prediction.

This report will cover the re-training of the locked model on the original dataset, as well as a full evaluation of the test dataset (including ROC curve, lift chart, and profit curve), actionable targeting strategies based on the data obtained, and a reflection of the methodology implemented across the checkpoint report and this report.

2.0 Test Set Evaluation

2.1 Model Application and Metrics

The locked Random Forest model was re-trained on the original 70% split to ensure that when it was fitted to the new unseen dataset, it would generalise as originally intended. Through exploratory analysis, it was found that the test dataset had a renewal rate of 10% (500 renewers from 5,000 customers), compared to the renewal rate of 6.25% that was seen in the original training set.

To determine the effectiveness of the model, a side-by-side comparison of performance on the validation set and the test set was conducted across various metrics (Table 1). The test AUC value observed confirms that the model retains a strong ability to discriminate on unseen data. It is worth noting that the sensitivity reported in Table 1 is measured at the default probability threshold of 0.5 and therefore does not directly reflect the campaign strategy. In actuality, the model ranks all 5,000 customers by their predicted renewal probability and targets the top-n, an approach independent of the threshold, as this provides better business value which will be evident in the lift and profit analysis presented in Sections 2.3 and 2.4.

Model	Accuracy	Sensitivity	Specificity	Precision	AUC
Validation Set	0.9571	0.3933	0.9947	0.8310	0.8903
Test Set	0.9598	0.6460	0.9947	0.9308	0.9325

Table 1. *Random Forest performance metrics on the validation set (checkpoint) and the test set. The highlighted row is the test set results.*

2.2 ROC Curve

The ROC curve illustrates the trade-off between sensitivity and specificity across all possible classification thresholds, with a curve closer to the top-left corner indicating stronger discrimination ability. Figure 1 presents the ROC curves for the Random Forest on both the test set (solid line) and the validation set (dashed line), plotted on the same axes. The purpose of showing both curves on the same axes is to allow for the confirmation of whether the AUC achieved on the validation set was a reliable estimate of model performance, as well as providing a visual assessment of how well the model generalises the new test data. From Figure 1, we can see that the test curve sits above the validation curve, indicating that the model transfers effectively to the unseen data and will provide a more reliable prediction for renewal intent.

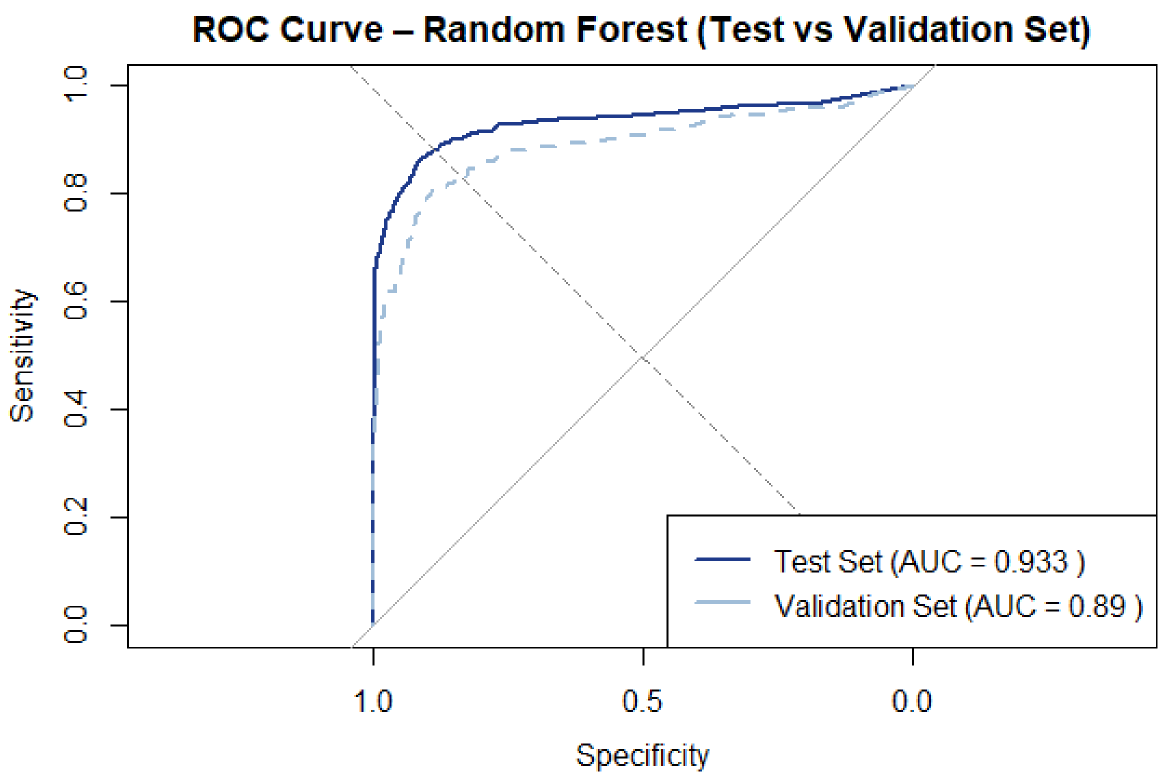

Figure 1. *ROC curves for the Random Forest on the test set (solid, dark blue) and validation set (dashed, light blue). AUC values are shown in the legend; the grey diagonal represents the random chance baseline (AUC = 0.5).*

2.3 Lift Chart

Inclusion of a lift chart provides a practical way to assess whether the model offers a genuine advantage over untargeted contact by comparing the renewal rate within a ranked group of customers against the overall renewal rate in the dataset. Figure 2 shows the cumulative lift chart for the Random Forest applied to the test set. A lift value of 1.0, shown by the horizontal dashed line, represents the performance of random selection, i.e. contacting customers with no guidance from the model. Any value observed above this threshold indicates the model is returning a higher proportion of renewers than chance alone would produce from the same number of contacts.

At the top 10% of ranked customers, equivalent to the 500-customer campaign budget, the model achieves a lift of 7.56. This means that within those 500 contracts, renewers are identified 7.56 times the rate that random selection would achieve. For Broughty Ferry Media, this directly translates into fewer wasted contacts and a substantially higher return on the £20,000 campaign investment than an untargeted approach would produce.

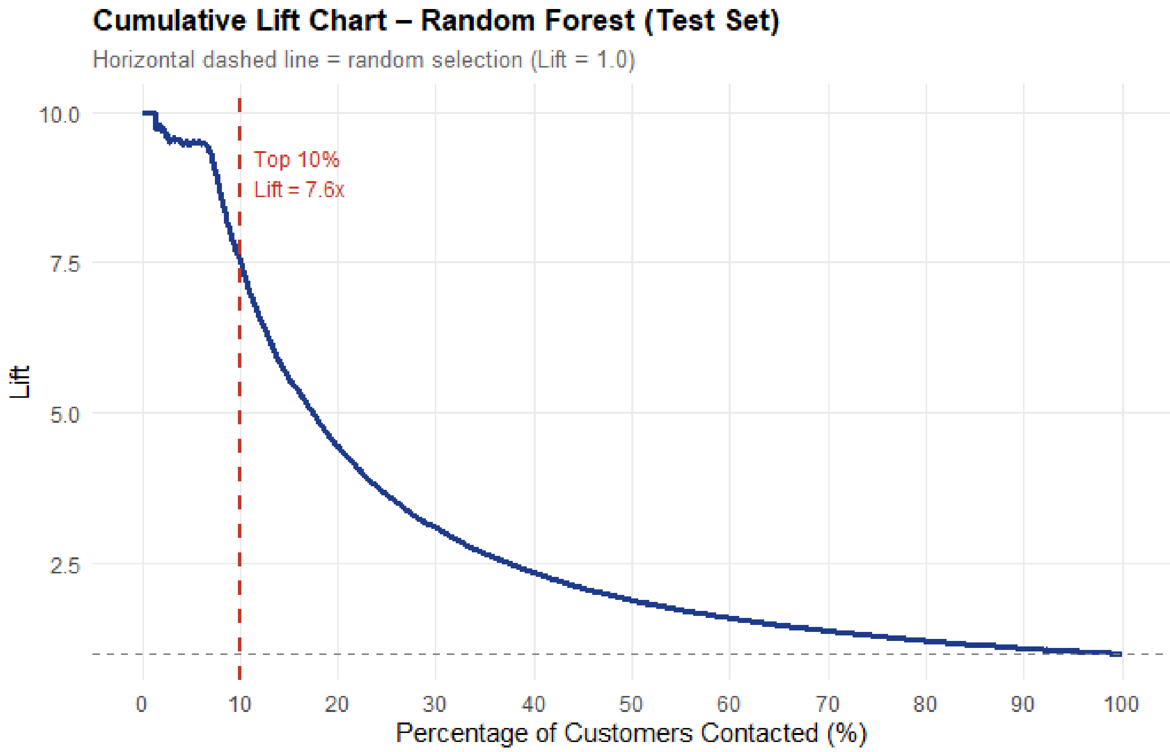

Figure 2. *Cumulative lift chart for the Random Forest on the test set. Values above 1.0 indicate the model captures more renewers than random selection. The red dashed line marks the top 10% (500 customers); Lift = ~7.56x at this threshold.*

2.4 Profit Curve

To determine how much profit the targeted campaign implementation would generate, a profit curve using the model's predictions was produced to visualise the cumulative profit generated at every targeting threshold. This is calculated as the revenue from actual renewals among the top-ranked customers, minus the total contact cost spent to reach them (Profit = (Renewals x £140) – (Contacts * £40)). Figure 3 presents this curve for the Random Forest across all its possible targeting thresholds on the test dataset. From Figure 3, it can be seen that the curve rises steeply in the top-500 customer section, reflecting the model’s strong concentration of genuine renewers at the highest predicted probabilities. The curve reaches its peak at an optimal threshold of 500 customers, beyond which each additional contact would most likely cost more than the renewal revenue is expected to generate.

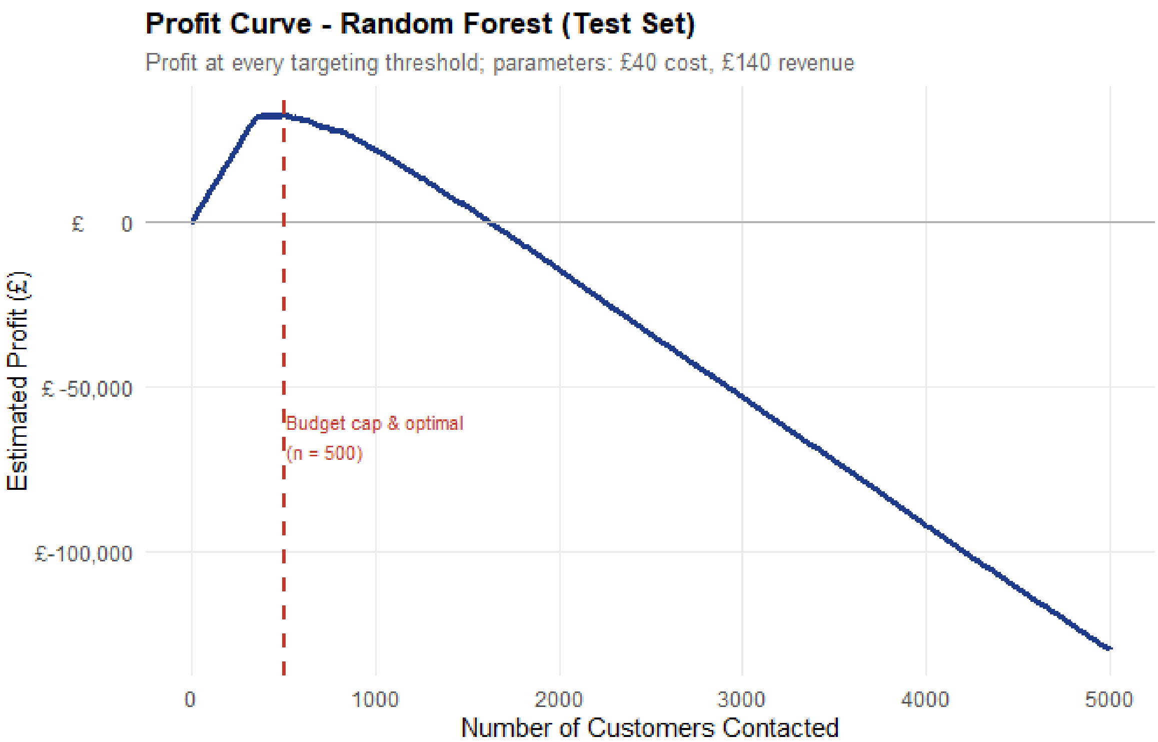

Figure 3. *Profit curve for the Random Forest across all targeting thresholds (test set). The red dashed line represents the budget cap and profit-maximising optimal threshold (500 customers). Parameters: £40 contact cost, £140 renewal revenue.*

3.0 Targeting Strategies

Three targeting strategies were evaluated on the test set to provide Broughty Ferry Media with a range of actionable options based on the various predictions observed from the 5,000 test customers (Table 2).

Strategy	Customers	Cost (£)	Renewals	Revenue (£)	Profit (£)
Random Selection (Baseline)	500	20,000	50	7,000	-13,000
Model – Top 500 (Full Budget)	500	20,000	378	52,920	32,920
Model – Optimal Threshold	500	20,000	378	52,920	32,920

Table 2. *Targeting strategy comparison on the test set. All monetary figures rounded. Highlighted rows = profit-maximising strategies. Baseline strategy assumes a 10% random renewal rate.*

3.1 Strategy 1 – Random Selection Baseline

After exploratory analysis showed that the test set's rate of renewal was 10%, it is expected that by targeting 500 customers at random, approximately 50 would renew, generating £7,000 in revenue against a £20,000 contact cost, resulting in a loss of £13,000. This baseline illustrates the cost of campaign implementation without the use of a predictive model.

3.2 Strategy 2 – Model-Guided Top 500 (Full Budget)

By targeting the 500 customers ranked highest for predicted renewal probability by the Random Forest model, the campaign would be expected to reach roughly 378 actual renewers (response rate of 75.6%). This strategy generates £52,920 (rounded up) in revenue from a £20,000 investment, producing a total profit of £32,920. This performance represents a substantial improvement when compared to the random baseline, confirming that the model’s rankings provide the business with meaningful value.

3.3. Strategy 3 – Optimal Threshold (Maximum Profit)

The profit curve presented in Figure 3 identifies the profit-maximising threshold as 500 customers, coinciding directly with the campaign budget cap. As the optimal threshold and full budget strategies produce the same outcomes of 378 renewals and a profit of £32,920, there is no trade-off between the implementation of either strategy. With this in mind, this report recommends that Broughty Ferry Media contact the top 500 customers ranked by the Random Forest model and deploy the full budget of £20,000 to generate the highest predicted profit.

4.0 Reflection

4.1 What Worked Well

The Random Forest model performed well on the test set, with PoundsPerIssue found to still be the most important predictor, as shown by the Mean Decrease in Gini Impurity. This finding is consistent with the variable importance results presented in the checkpoint report. The decision to calculate the point-biserial correlations of predictors on the training data and confirm them through VIF testing allowed for the production of ten variables that were compact and non-redundant. When creating a framework to evaluate performance, combining AUC and profit simulation proved to be beneficial as both measures supported the same model selection choice. This decision was then further reinforced by the profit curve, as this confirmed that the optimal threshold aligns with the full campaign budget.

4.2 What Did Not Work as Well

In the variable selection process, categorical variables such as MagazineStatus, LastPaymentType, and GiftDonor were omitted because point-biserial correlation is restricted to numeric predictors. This was a result of the checkpoint report seeking to maintain a singular correlation methodology for the sake of consistency. Looking back, employing dummy encoding on these predictors before selection might have facilitated the identification of additional predictors of renewal intent, representing a clear opportunity for enhancement in subsequent iterations.

Furthermore, the decision tree demonstrated superior sensitivity compared to the Random Forest at the default 0.5 probability threshold, achieving a sensitivity of 0.4867, as opposed to 0.3933 in the checkpoint report. This observation could imply that, although the Random Forest is the more robust model for ranking customers based on renewal probability, it may under-classify renewers when a binary prediction is necessitated. Consequently, for future campaigns, modifying the probability cutoff from 0.5 to 0.3 could potentially recover some of the sensitivity lost without adversely affecting the model’s AUC performance.

4.3 Why Test Performance May Differ from Validation

The validation set was drawn from the same 8,000-row population as the training data using stratified sampling, and as it was used to compare and select between models, the validation AUC reflects performance on Data that informed the decision-making process. This means that the validation estimate may be slightly optimistic when compared to the performance on a dataset that had no influence whatsoever. The test set satisfies this condition as it had no role in the model decision-making process, allowing it to provide a more objective measure of predictive performance.

A further consideration is the difference in renewal rates between the training set and test set, with the training having a rate of 6.25%, and the test set having a rate of 10%. This higher concentration of renewers increases the signals available to the model and may partially account for any improvement in AUC relative to the estimate. As this gap in renewal rates shows, model performance is dependent on shifts in the composition of the customer population. This suggests that before implementation on Broughty Ferry Media’s database, they will need to monitor the renewal rates of future cohorts, as they may fluctuate and impact the predictive capability of the model.

5.0 Conclusion

This report continued the work of the checkpoint report by applying the locked Random Forest model to the test dataset, ultimately confirming its strong discriminative and predictive ability. The data collected from this model was then used to shape clear and actionable business recommendations, with the key recommendation suggesting that for maximum and optimum profit generation, the full budget targeting the highest-ranked 500 customers should be employed, generating £32,920 in profit.

The foundations that were established in both the checkpoint report and final report, consisting of stratified splitting, data-driven selection, training of various models, and business-aligned evaluation, provide Broughty Ferry Media with a robust framework that can be applied to their future campaign cycles. After review of the methods used in these reports, the recommended next steps should focus on including the categorical predictors that were omitted from variable selection, as well as exploring what impact changes in threshold calibration would have, and monitoring of renewal rate shifts between cohorts to maintain model viability.